Medical Insurance Cost Prediction

1. Introduction

Healthcare expenses are increasing rapidly nowadays. Insurance companies need to predict medical costs in advance to decide suitable premiums for customers.

Manual calculation of insurance charges is difficult and time-consuming. Therefore, Machine Learning techniques can be used to predict medical insurance costs automatically based on personal details such as age, BMI, smoking habits, gender, number of children, and region.

In this project, a Linear Regression model is built to predict medical insurance charges accurately.

2. Objectives

The main objectives of this project are:

- To analyze the insurance dataset
- To perform data preprocessing
- To visualize the data using graphs
- To build a Linear Regression model
- To predict insurance charges
- To evaluate model accuracy

3. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Google Colab

4. Dataset Description

The dataset used is insurance.csv.

Total records: 1338
Total columns: 7

Features:

- age
- sex
- bmi
- children
- smoker
- region
- charges (target)

Charges represent the medical insurance cost to be predicted.

5. Data Loading and Understanding

The dataset was loaded into a Pandas DataFrame.
The first few rows and dataset information were checked to understand the structure.

Basic operations performed:

- head()
- shape
- info()

```
# getting some informations about the dataset
insurance_dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
```

```
# number of rows and columns
insurance_dataset.shape
```

```
(1338, 7)
```

```
# first 5 rows of the dataframe
insurance_dataset.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

6. Exploratory Data Analysis (EDA)

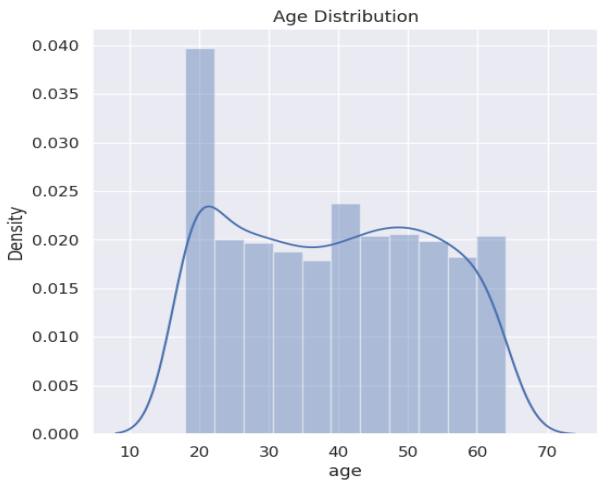
Data visualization was performed to understand patterns and relationships.

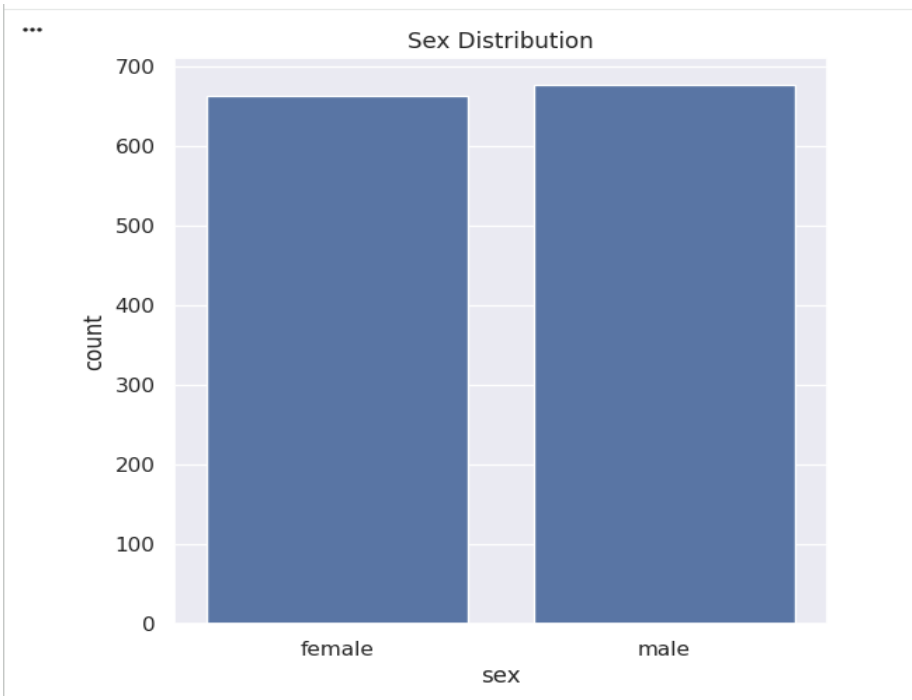
The following graphs were plotted:

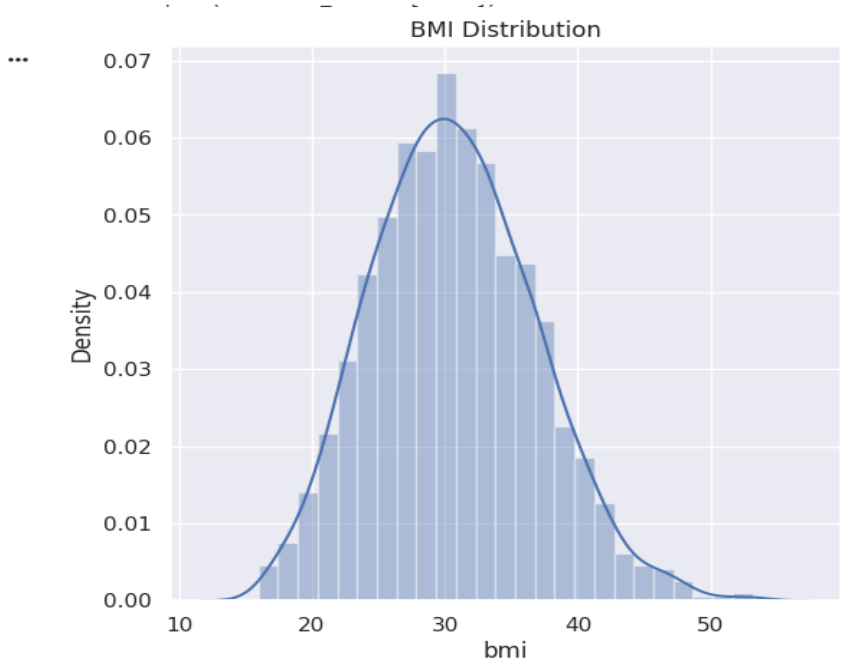
- Age distribution
- Gender distribution
- BMI distribution
- Children count
- Smoker count
- Region count
- Charges distribution

Observations:

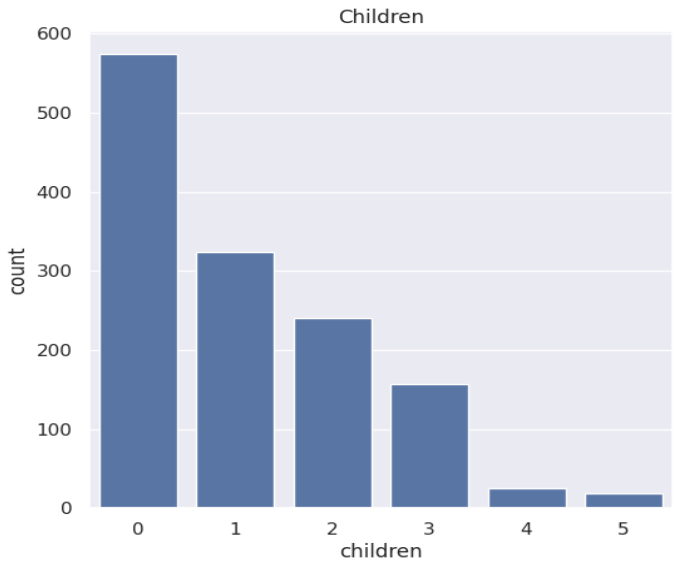
- Smokers have higher charges
- Higher BMI increases medical cost
- Middle-aged people pay more insurance

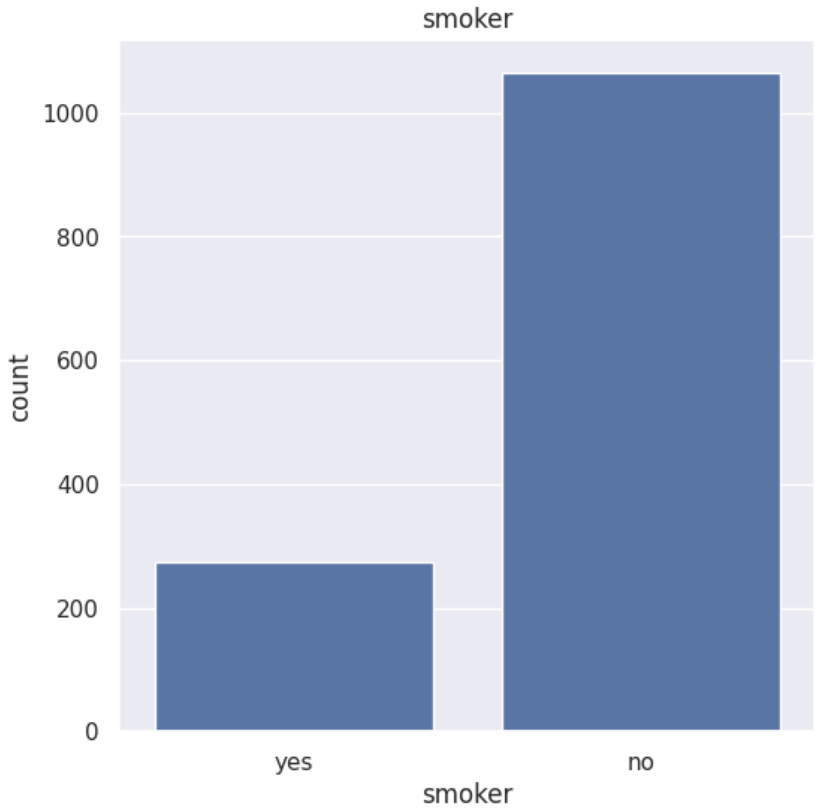






Normal BMI Range --> 18.5 to 24.9

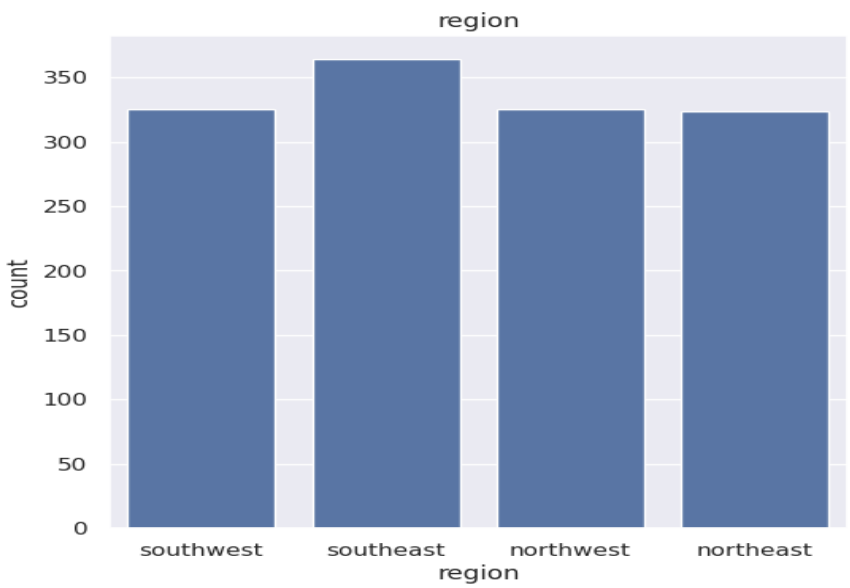


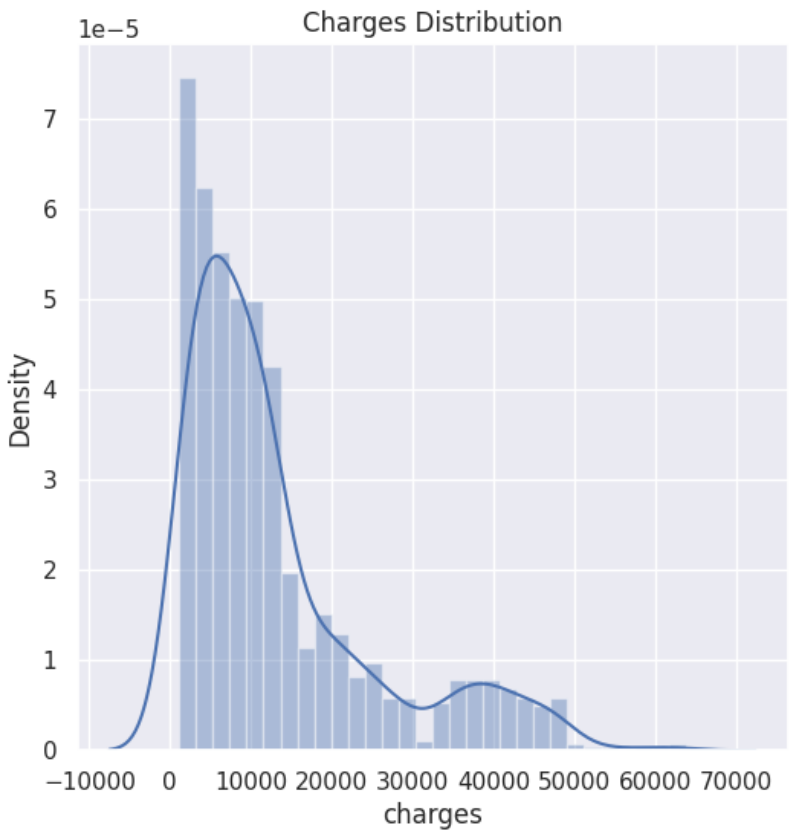



```
insurance_dataset['smoker'].value_counts()
```

	count
smoker	
no	1064
yes	274

dtype: int64


region
count
350
300
250
200
150
100
50
0
southwest
southeast
northwest
northeast
region


Charges Distribution
1e−5
Density
7
6
5
4
3
2
1
0
−10000
0
10000
20000
30000
40000
50000
60000
70000
charges

7. Data Preprocessing

Machine learning models require numerical data.

So categorical features were encoded:

- male → 0, female → 1
- yes → 0, no → 1
- region converted to numbers

No missing values were found.

```
# checking for missing values
insurance_dataset.isnull().sum()
```

	0
age	0
sex	0
bmi	0
children	0
smoker	0
region	0
charges	0

dtype: int64

8. Feature and Target Selection

Features (X):

- age, sex, bmi, children, smoker, region

Target (Y):

- charges

Features are used to predict the insurance cost.

9. Train-Test Split

The dataset was divided into:

- 80% Training data

- 20% Testing data

Training data is used to train the model and testing data is used to check accuracy.

10. Model Used – Linear Regression

Linear Regression is a supervised learning algorithm used to predict continuous values.

It is simple, fast, and suitable for cost prediction problems.

The model learns the relationship between input features and insurance charges.

11. Model Training

The Linear Regression model was trained using training data.

Steps:

- Model created
- Model fitted on training data

The model learns patterns from the dataset.

12. Model Evaluation

Model accuracy was evaluated using R^2 score.

Results:

- Training accuracy ≈ 0.75
- Testing accuracy ≈ 0.74

This shows the model predicts insurance costs with good accuracy.

```python
print(X)
```

```
...        age  sex     bmi  children  smoker  region
0       19    1  27.900         0       0       1
1       18    0  33.770         1       1       0
2       28    0  33.000         3       1       0
3       33    0  22.705         0       1       3
4       32    0  28.880         0       1       3
...    ...  ...     ...       ...     ...     ...
1333    50    0  30.970         3       1       3
1334    18    1  31.920         0       1       2
1335    18    1  36.850         0       1       0
1336    21    1  25.800         0       1       1
1337    61    1  29.070         0       0       3

[1338 rows x 6 columns]
```

```python
print(Y)
```

```
0       16884.92400
1        1725.55230
2        4449.46200
3       21984.47061
4        3866.85520
            ...
1333    10600.54830
1334     2205.98080
1335     1629.83350
1336     2007.94500
1337    29141.36030
Name: charges, Length: 1338, dtype: float64
```

```python
# R squared value
r2_train = metrics.r2_score(Y_train, training_data_prediction)
print('R squared vale : ', r2_train)
```

```
R squared vale :  0.751505643411174
```

```python
# prediction on test data
test_data_prediction =regressor.predict(X_test)
```

```python
# R squared value
r2_test = metrics.r2_score(Y_test, test_data_prediction)
print('R squared vale : ', r2_test)
```

```
R squared vale :  0.7447273869684076
```

13. Prediction System

A predictive system was created to predict insurance cost for new user input.

The model successfully predicted the medical charges.

Example prediction was generated using sample values.

```
input_data = (31,1,25.74,0,1,0)

# changing input_data to a numpy array
input_data_as_numpy_array = np.asarray(input_data)

# reshape the array
input_data_reshaped = input_data_as_numpy_array.reshape(1,-1)

prediction = regressor.predict(input_data_reshaped)
print(prediction)

print('The insurance cost is USD ', prediction[0])
```

```
[3760.0805765]
```

14. Results

- Data successfully analyzed
- Model trained successfully
- Good accuracy achieved
- Insurance charges predicted correctly
- System works for new data

15. Applications

This project can be used in:

- Insurance companies
- Hospitals
- Healthcare analytics
- Premium calculation systems

16. Conclusion

This project demonstrates how Machine Learning can be used to predict medical insurance costs effectively.

Linear Regression provided good accuracy and reliable predictions. The system reduces manual effort and helps insurance companies make better decisions.

Thus, the project successfully achieves its objectives.

17. Future Scope

- Use advanced algorithms (Random Forest, XGBoost)
- Increase dataset size
- Build web application
- Improve prediction accuracy